

```
public class StringCharacterMethods {  
    public static void main(String[] args) {  
        // ===== STRING METHODS  
        =====  
  
        String str = " Hello World! ";  
        String str2 = "hello";  
  
        // Length & Character Access  
        System.out.println("Length: " + str.length());  
        System.out.println("Char at index 1: " + str.charAt(1));  
        System.out.println("To Char Array: " +  
java.util.Arrays.toString(str.toCharArray()));  
  
        // Comparison  
        System.out.println("Equals: " + str.equals(str2));  
        System.out.println("Equals Ignore Case: " + str.equalsIgnoreCase(str2));  
        System.out.println("Compare To: " + str.compareTo(str2));  
  
        // Searching  
        System.out.println("Contains 'World': " + str.contains("World"));  
        System.out.println("Index of 'o': " + str.indexOf('o'));  
        System.out.println("Last Index of 'o': " + str.lastIndexOf('o'));  
  
        // Modification  
        System.out.println("Concatenation: " + str.concat(" Java"));  
        System.out.println("Replace 'o' with 'a': " + str.replace('o', 'a'));  
        System.out.println("Substring (1-5): " + str.substring(1, 5));  
    }  
}
```

```
System.out.println("Trim: " + str.trim());
```

```
System.out.println("Repeat (3 times): " + str.repeat(3));
```

```
// Case Conversion
```

```
System.out.println("To Uppercase: " + str.toUpperCase());
```

```
System.out.println("To Lowercase: " + str.toLowerCase());
```

```
// Splitting & Joining
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```
String[] parts = str.split(" ");
```

```
System.out.println("Split by space: " + java.util.Arrays.toString(parts));
```

```
System.out.println("Join with '-': " + String.join("-", parts));
```

```
// Conversion
```

```
System.out.println("Value of 100: " + String.valueOf(100));
```

```
System.out.println("Formatted: " + String.format("Age: %d", 25));
```

```
// Checking Empty or Blank (Java 11+)
```

```
System.out.println("Is Empty: " + str.isEmpty());
```

```
System.out.println("Is Blank: " + str.isBlank());
```

```
// Matching
```

```
System.out.println("Matches digits: " + "12345".matches("\\d+"));
```

```
// ===== CHARACTER METHODS
```

```
=====
```

```
char ch = 'A';
```

```
// Checking Properties
System.out.println("Is Letter: " + Character.isLetter(ch));
System.out.println("Is Digit: " + Character.isDigit(ch));
System.out.println("Is Letter or Digit: " + Character.isLetterOrDigit(ch));
System.out.println("Is Lowercase: " + Character.isLowerCase(ch));
System.out.println("Is Uppercase: " + Character.isUpperCase(ch));
System.out.println("Is Whitespace: " + Character.isWhitespace(' '));

// Conversion
System.out.println("To Lowercase: " + Character.toLowerCase(ch));
System.out.println("To Uppercase: " + Character.toUpperCase('b'));
System.out.println("Numeric Value: " + Character.getNumericValue('9'));

// Unicode & Advanced Checks
System.out.println("Is Alphabetic: " + Character.isAlphabetic(ch));
System.out.println("Is Defined: " + Character.isDefined(ch));
System.out.println("Is ISO Control: " + Character.isISOControl('\n'));

// Comparison
System.out.println("Character Compare: " + Character.compare('a', 'b'));
}
}
```